

Anshul Israni

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TECHNICAL SKILLS

- **Programming:** C, Python, Verilog, ARM Assembly
- **Hardware Design & Simulation:** Altium Designer, KiCAD, CircuitMaker, Cadence, PSIM
- **Embedded Systems:** ESP32, STM32, ATmega328P, PIC32, Arduino; UART, SPI, I²C, CAN Bus; Bluetooth Low Energy (BLE)
- **Modelling & FPGA:** MATLAB/Simulink; ModelSim, Quartus
- **Prototyping & Fabrication:** PCB Layout & Assembly, Soldering, 3D Printing, SolidWorks

EDUCATION

University of British Columbia

September 2020 - May 2026

Bachelor of Applied Science - Electrical Engineering, APSC Co-op Program

- **Relevant Coursework:** Power Electronics, Control Systems, Analog CMOS IC Design, Digital Systems Design, VLSI Systems
- **Awards:** Dean's Honors List (2024-25), Outstanding International Student Scholarship (\$15K CAD)

WORK EXPERIENCE

System Control and Energy Management Co-op

May 2025 - April 2026

Schneider Electric (Solar and Storage Division)

Richmond, BC

- Developed **system-level logic** for 8+ power-flow modes within the Real Time Power Controller (RTPC) in **MATLAB/Simulink**, ensuring grid-compliant behavior and preventing inverter import/export limit violations
- Authored RTPC power-flow validation cases in Jama and automated their execution in **MATLAB/Simulink**, verifying state transitions and correct source prioritization
- Worked with senior engineers to perform **Hardware-in-the-Loop (HIL) testing**, correlating simulated behaviors with physical RTPC responses and identifying mismatches in battery SoC transitions and solar availability logic

Electrical & Electronics Engineer Co-op

January 2025 - March 2025

Fingerprint Technologies (Startup developing e-bike solutions)

North Vancouver, BC

- Implemented **BLE firmware** on ESP32 in **embedded C** by defining custom GATT characteristics for speed and cadence, enabling successful device discovery and real-time telemetry streaming to a mobile application
- Performed **hardware bring-up of motor control PCBs** by resolving 5+ soldering defects and component population errors, restoring stable motor drive performance, validated via VESC
- Conducted **motor thermal performance** testing using a VESC-integrated 2WD dyno rig under variable load conditions, and identified optimal speed ranges across 10+ gradients of a hilly terrain

Battery & Control-Circuitry Team Lead

May 2022 - December 2024

UBC Thunderbikes (Electric motorcycle student design team)

Vancouver, BC

- Led a team of 5 in designing and integrating a **100V LiFePO₄ battery system** for an in-house custom-built electric motorcycle
- Designed and fabricated an I/O wire harness to interface the battery pack with the Battery Management System (BMS), enabling real-time SoC / SoH monitoring via **CAN communication** for proactive battery protection
- Implemented a 102.4V / 7.5A **CC-CV charging circuit** with voltage sensing and current regulation, interfacing with the BMS to ensure safe and efficient battery charging
- Tested and debugged relay-control **embedded firmware in C** using STM32CubeIDE, verifying safe state transitions between ignition, operation, and charging modes

Scientific Data Management Co-op

September 2023 - April 2024

STEMCELL Technologies (R&D Operations Division)

Vancouver, BC

- Implemented an inventory management system to **automate data collection** for 1000+ lab assets, reducing search time by 25%

PROJECTS

Phased Array Antenna System

September 2024 - April 2025

- Designed a 2×2 phased array to track a live laboratory transmitter, serving as a PoC for real-time satellite tracking applications
- Simulated a 20×20 phased array in **MATLAB** using MUSIC-based DOA estimation and Doppler shift modeling to validate digital beamforming scalability beyond a 2×2 prototype
- Implemented **FPGA** beamforming modules, including real-time IQ phase shifting and trigonometric LUTs, using **SystemVerilog** to achieve electronic beam steering, demonstrating the feasibility of hardware-accelerated beamforming for real-time satellite tracking
- Engineered a lightweight, modular **3D-printed** PLA frame, ensuring accurate antenna alignment and environmental durability

Robotic Gripper System

January 2023 - April 2023

- Designed a gripper mechanism to enable precise object manipulation through detailed modeling and system optimization, achieving stable closed-loop position control with < 5% steady-state error using **PID tuning in MATLAB and Simulink**
- Improved motor responsiveness and system stability by implementing a quadrature decoder for precise position tracking, an **H-bridge** for bidirectional control, and a **9V-5V voltage regulator** for efficient power management
- Created detailed models using **SolidWorks** and conducted finite element analysis to ensure structural integrity and performance

Coin Picking Autonomous Robot

January 2022 - April 2022

- Designed an autonomous mobile robot capable of detecting, identifying, and collecting coins within a defined perimeter
- Implemented a discrete **MOSFET-based H-bridge** with optocoupler isolation for bidirectional DC motor control of the drive system
- Integrated **PWM-based servo actuation** into the control architecture, enabling precise electromagnet positioning for coin placement
- Developed an LC tank circuit for perimeter detection and a Colpitts oscillator-based metal detector, identifying coins through measurable frequency shifts